Exact second-order strong-coupling expansion of the spectral gap of SU(2) Kogut–Susskind lattice gauge theory on the three-torus

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Abstract

We compute the first two nontrivial orders of the strong-coupling expansion of the spectral gap of the pure SU(2) Kogut–Susskind Hamiltonian on the periodic three-torus $\mathbb{T}^3_L$ of side L, in the dimensionless normalization $H(\mu) = H_E + \mu H_M$ with $H_E = \sum_\ell C_2(\ell)$ and $H_M = \sum_p (2 - \Re \operatorname{Tr} U_p)$; the dictionary to the Kogut–Susskind coupling at unit lattice spacing is $\mu = 2g^{-6}$, so that in physical units $\Delta_{\mathrm{phys}}(g, L) = \frac{3}{2} g^2 - \frac{142}{105}\, g^{-10} + O(g^{-16})$. For every $L \geq 5$ the gap above the vacuum obeys

$$\Delta(\mu, L) \;=\; 3 - \frac{71}{105}\,\mu^2 + O_L(\mu^3), \qquad \mu \to 0^+,$$

with an L-independent coefficient; the band head is the translation-invariant one-plaquette excitation. At $L = 4$ an additional winding sector enters the $E = 3$ eigenspace and the gap is governed by that sector with coefficient at most $-16/21$. All group-theoretic coefficients are exact rational numbers derived from Peter–Weyl orthogonality, $6j$ recoupling, and the SU(2) Weingarten calculus; every step is machine-verifiable in exact rational arithmetic. The statement is a fixed-volume perturbative theorem; no uniformity in L and no continuum limit are claimed.

1 Introduction and result

Strong-coupling expansions of Hamiltonian lattice gauge theory go back to Kogut and Susskind [1] and to the glueball calculations of Kogut, Sinclair and Susskind [2]; textbook accounts appear in [4, 3]. Within that framework we obtain an exact coefficient of the second-order expansion of the spectral gap, together with a complete finite-volume band analysis, in a normalization fixed once and for all from the physical Hilbert space. The exact rational value and the finite-volume band-head statement are the objects of interest; their status relative to the published literature is discussed in Section 9 and in the accompanying novelty ledger.

Let $\Lambda_L = (\mathbb{Z}/L\mathbb{Z})^3$, $L \geq 4$, with link set E and plaquette set $\mathcal{P}$ ($|\mathcal{P}| = 3L^3$). The physical Hilbert space is $\mathcal{H}_0 = \{f \in L^2(\mathrm{SU}(2)^E, dU)\}$, the gauge-invariant subspace, dU the product Haar measure. The Hamiltonian is

$$H(\mu) \;=\; H_E + \mu H_M, \qquad H_E = \sum_{\ell \in E} C_2(\ell), \qquad H_M = \sum_{p \in \mathcal{P}} \big(2 - \Re \operatorname{Tr} U_p\big), \tag{1}$$

where $C_2(\ell)$ is the (left) Casimir on link ℓ, acting as $j(j+1)$ on the Peter–Weyl component of spin j, and U_p is the ordered holonomy around plaquette p. $H_M \geq 0$, $H_E \geq 0$; $H(\mu)$ is self-adjoint, bounded below, with discrete spectrum at fixed L (H_E has compact resolvent, H_M is bounded). The vacuum $\Omega \equiv 1$ has $H(0)\Omega = 0$; for $\mu > 0$ small it is the unique ground state (analytic perturbation theory, Section 3).

Theorem 1.1 (main)**.** *For every fixed $L \geq 5$ there exists $\mu_0(L) > 0$ such that for $0 < \mu < \mu_0(L)$ the spectral gap of $H(\mu)$ above its ground state is*

$$\Delta(\mu, L) \;=\; 3 - \frac{71}{105}\,\mu^2 + O_L(\mu^3), \qquad \mu \to 0^+.$$

The coefficient $-71/105$ is independent of L. The second-order band minimum is attained at crystal momentum $k = 0$, in the uniform orientation-symmetric one-plaquette mode, uniquely among band minima at second order.

Corollary 1.2 (physical units)**.** *With $H_{\mathrm{KS}} = \frac{g^2}{2}\sum_\ell E^2(\ell) - \frac{1}{2g^4}\sum_p(\operatorname{Tr} U_p + \operatorname{Tr} U_p^\dagger)$ at $a = 1$ and $L \geq 5$,*

$$\Delta_{\mathrm{phys}}(g, L) = \frac{g^2}{2}\,\Delta\big(2g^{-6}, L\big) = \frac{3}{2}\,g^2 - \frac{142}{105}\,g^{-10} + O_L(g^{-16}) = \frac{3}{2}\,g^2\Big(1 - \frac{284}{315}\,g^{-12} + O_L(g^{-18})\Big), \qquad g \to \infty.$$

Every conversion factor is exact rational; a source quoting $\mu = g^{-4}$ for the same operator differs by the factor $c = 2$ and must be rescaled by $c_2 \mapsto c_2/c^2$ before comparison.

Theorem 1.3 ($L = 4$ exception)**.** *At $L = 4$ the $E = 3$ eigenspace of H_E contains, besides the 192 plaquette states, 48 straight winding-loop states. The spectral gap is then governed by the winding sector, whose second-order coefficient satisfies $c_2(L{=}4) \leq -16/21 < -71/105$; the exact value of $c_2(L{=}4)$ is not determined in this paper (Section 8).*

Both theorems are fixed-volume statements. They say nothing about $\inf_L \Delta(\mu, L)$ uniformity, and nothing about the continuum limit. Section 9 states the precise scope boundary.

2 Setup, conventions, and spectral preliminaries

2.1 Hilbert space and gauge invariance

$\mathcal{H} = L^2(\mathrm{SU}(2)^E, dU)$ with $dU = \prod_\ell dU_\ell$ the product of Haar measures normalized to 1. Gauge transformations act by $(g \cdot f)(U) = f(\{\gamma_\ell^{-1} U_\ell \gamma_{\ell'}\})$ for vertex functions $\gamma_x \in \mathrm{SU}(2)$; $\mathcal{H}_0$ is the invariant subspace. In the Peter–Weyl decomposition

$$L^2(\mathrm{SU}(2)) \;\cong\; \widehat{\bigoplus}_{j \in \frac{1}{2}\mathbb{N}} V_j \otimes V_j^*, \qquad C_2\big|_{V_j} = j(j+1),$$

gauge invariance means: at each vertex x, the tensor product of the V_j's carried by links incident to x contains the singlet, and the state is the corresponding intertwiner contraction. A basis of $\mathcal{H}_0$ is labelled by flux assignments $\{j_\ell\}$ plus intertwiners at vertices; the H_E-eigenvalue is $\sum_\ell j_\ell(j_\ell + 1)$.

2.2 Normalization

The primary object of this paper is the dimensionless family (1) with $\mu > 0$ as the expansion parameter; $C_2(\ell) = E^2(\ell)$ on class functions. Dictionary to the Kogut–Susskind Hamiltonian $H_{\mathrm{KS}} = \frac{g^2}{2}\sum_\ell E^2(\ell) - \frac{1}{2g^4}\sum_p(\operatorname{Tr} U_p + \operatorname{Tr} U_p^\dagger)$ at $a = 1$: in units of $g^2/2$,

$$\frac{2}{g^2}H_{\mathrm{KS}} + \frac{4|\mathcal{P}|}{g^6} = \sum_\ell C_2(\ell) + \frac{2}{g^6}\sum_p\big(2 - \Re\operatorname{Tr} U_p\big), \qquad \text{i.e.} \quad \mu = \frac{2}{g^6}.$$

Some references (and the earlier notes of this campaign) absorb the factor of 2 into the coupling and quote $\mu = g^{-4}$ under the equally standard convention $H_B = -g^{-4}\sum_p \Re\operatorname{Tr} U_p$. All results below are statements about $H(\mu)$ itself; a change of convention $\mu \mapsto c\,\mu$ rescales the coefficient as $c_2 \mapsto c_2/c^2$. Conversion factors to $\lambda = \frac{1}{2}g^{-4}$-type variables are tabulated in Appendix D.

2.3 Analyticity

H_M is bounded ($\|\Re\operatorname{Tr} U_p\|_\infty \le 2$), so $H(\mu)$ is a bounded analytic family for $\mu \in \mathbb{R}$ in the norm-resolvent sense at fixed L. The spectrum is discrete; simple eigenvalues and isolated clusters admit convergent power series in μ (Kato). The vacuum cluster and the $E = 3$ cluster of $H(0)$ are separated by 3, so for μ small the gap is the bottom of the deformed $E = 3$ cluster minus the deformed ground energy, each analytic in μ on the real axis (the $E = 3$ cluster is degenerate; its splitting is handled by degenerate perturbation theory, which yields the expansions of Theorem 1.1 with $O_L(\mu^3)$ remainder).

3 The $E = 3$ eigenspace and Wilson-loop orthonormality

Lemma 3.1 (flux classification)**.** *The $E = 3$ eigenspace of H_E on $\mathcal{H}_0$ is spanned by gauge-invariant states carrying flux $j = \frac{1}{2}$ on exactly four links and 0 elsewhere. For $L \ge 5$ the supporting links form a plaquette; for $L = 4$ they form a plaquette or a straight winding loop of length 4.*

Proof. $3 = \sum_\ell j_\ell(j_\ell+1)$ with each $j_\ell(j_\ell+1) \in \{0, \frac{3}{4}, 2, \frac{15}{4}, \dots\}$ forces four links with $j = \frac{1}{2}$ (since $2 + \frac{3}{4} < 3$ and $\frac{15}{4} > 3$, and $3/4$ does not divide the remainder 1 otherwise). Gauge invariance requires the total flux at each vertex to contain a singlet, so each vertex carries an even number (0 or 2, as the j's are $\frac{1}{2}$) of flux links: the support is a closed loop of length 4. On $\mathbb{T}^3_L$ a closed nearest-neighbor loop of length 4 is a plaquette, except when $L = 4$ and the loop wraps once in a coordinate direction (a straight winding loop). □

Lemma 3.2 (singlet uniqueness and the Wilson-loop basis)**.** *For a fixed closed loop $C = (\ell_1, \dots, \ell_4)$ with $j = \frac{1}{2}$ on each link, the space of gauge-invariant tensors is one-dimensional, spanned by the Wilson loop function $\chi_{1/2}(U_C) = \operatorname{Tr}(U_{\ell_1}\cdots U_{\ell_4})$.*

Proof. Each interior vertex of the loop pairs exactly two $V_{1/2}$ factors, whose singlet space is one-dimensional ($V_{1/2} \otimes V_{1/2} \supset V_0$ with multiplicity one, via the invariant $\varepsilon \in V_{1/2} \otimes V_{1/2}$). The product of the four vertex singlets, propagated around the loop, is the cyclic index contraction $\operatorname{Tr}(U_{\ell_1}\cdots U_{\ell_4})$; uniqueness of the singlet at each vertex forces any intertwiner to be a scalar multiple of it. □

Lemma 3.3 (orthonormality)**.** *For loops C, C' of length 4 as in Lemma 3.1,*

$$\big\langle \chi(U_C), \chi(U_{C'}) \big\rangle_{L^2(\mathrm{SU}(2)^E)} = \delta_{CC'}.$$

Proof. $\overline{\chi(U_{C'})} = \chi(U_{C'}^\dagger)$, so the inner product is the integral of a product carrying one U-type factor from $\chi(U_C)$ and, on C', one $\bar{U}$-type factor from $\overline{\chi(U_{C'})}$. If $C \neq C'$, some link $\ell \in C \setminus C'$ carries a single U-type factor and no $\bar{U}$-type factor; the remaining integrations produce a free matrix index on ℓ, and $\int U^a{}_b(U_\ell)\, dU_\ell = 0$ because the spin-$\frac{1}{2}$ matrix coefficient has no invariant vector. Hence the integral vanishes. If $C = C'$, Peter–Weyl orthogonality $\int U^a{}_b\, \overline{U^c{}_d}\, dU = \frac{1}{2}\delta^a_d \delta^c_b$ is applied link by link: each of the four links contributes a factor $\frac{1}{2}$ and each of the four successive index sums a factor 2, so the closed loop evaluates to $2^4 \cdot 2^{-4} = 1$. The full index chain is spelled out in Appendix A and machine-verified by two independent exact methods (Weyl integration and Weingarten) in `verify_anchors.py` [6]. □

The proof of the diagonal case is spelled out with all index positions in Appendix A; it is also machine-verified by two independent exact methods (Weyl integration and Weingarten) in `verify_anchors.py` [6].

Definition 3.4. *For $L \geq 5$ let $|p\rangle := \chi_{1/2}(U_p)$, $p \in \mathcal{P}$, the orthonormal basis of the $E = 3$ eigenspace $\mathcal{M}$; P denotes the orthogonal projection onto $\mathcal{M}$. For $L = 4$ add winding states $|w\rangle$.*

4 First order

Write $V := H_M - 2|\mathcal{P}|\,\mathbf{1} = -\sum_q \chi(U_q)$; $H(\mu) = H_E + 2|\mathcal{P}|\mu + \mu V$.

Lemma 4.1 (first-order cancellation)**.** *$PVP = 0$ on $\mathcal{M}$, and $\langle\Omega|V|\Omega\rangle = 0$; moreover $\langle p|V|\Omega\rangle = -\delta_{pq}$ summed, i.e. $\langle p|V|\Omega\rangle = -1$ for each p.*

Proof. $\langle p|V|q\rangle = -\sum_r \int \chi(U_p)\,\chi(U_r)\,\overline{\chi(U_q)}\, dU$. The integrand has total degree $(2,1)$ in the fundamental matrix coefficients (one $\bar{U}$-type factor from $\overline{\chi(U_q)}$, two U-type). Under the center element $z = -\mathbf{1} \in \mathrm{SU}(2)$ the integrand acquires a factor $z^2\bar{z} = z$ (i.e. -1), while the measure is invariant; hence every integral vanishes. The same parity argument kills $\langle\Omega|V|\Omega\rangle$ (degree $(1,0)$). For $\langle p|V|\Omega\rangle = -\sum_r \int \chi(U_p)\overline{\chi(U_r)}\, dU = -\sum_r \delta_{pr} = -1$ by Lemma 3.3. □

Thus the first-order correction to the band is exactly 0, and the first correction to the ground energy is second order (Section 6).

5 Second-order channels

Exhaustion of the intermediate spectrum — that the four channels below are *all* the intermediate states with nonzero matrix elements — is not assumed; it is proved in Appendix F, Proposition F.4, from the Peter–Weyl selection rules and a lattice-graph lemma. The second-order effective operator on $\mathcal{M}$ is

$$M_2 \;=\; P\,V\,Q\,(3 - H_E)^{-1} Q\,V\,P,$$

where $Q = 1 - P$ and $(3 - H_E)^{-1}$ acts on $Q\mathcal{H}_0$ (intermediate states have $E_n \geq 4 \cdot \frac{1}{2}(\frac{1}{2} + 1) = 3$, and $E_n = 3$ states lie in $\mathcal{M}$; the resolvent is negative). We classify the intermediate states contributing to $\langle p|M_2|q\rangle$ by the support of the inserted $\chi(U_r)$ relative to p, q.

5.1 Vacuum intermediate

$|\Omega\rangle\langle\Omega|$ contributes $\frac{\langle p|V|\Omega\rangle\langle\Omega|V|q\rangle}{3-0} = \frac{1}{3}$ for every pair (p,q), i.e. the operator $+\frac{1}{3}J$ with J the all-ones matrix in the $|p\rangle$ basis.

5.2 Single-plaquette (χ_1) intermediate, $p = q$

With $r = p$ the insertion acts on $|p\rangle$ itself. Using the pointwise fusion identity for SU(2) characters, $\chi_{1/2}(g)^2 = \chi_0(g) + \chi_1(g)$,

$$-\chi(U_p)^2 \;=\; -\mathbf{1} \;-\; \chi_1(U_p), \qquad \chi_1(U_p) := \mathrm{Tr}_{\mathrm{sym}^2}\big(U_{\ell_1}\cdots U_{\ell_4}\big).$$

The constant term is the vacuum intermediate of Section 5.1. The state $|\chi_1(p)\rangle := \chi_1(U_p)$ is gauge-invariant, carries pure flux $j = 1$ on each of the four links of p (each link appears once, in the spin-1 representation), hence has $E_n = 4 \cdot 1 \cdot 2 = 8$ and, by the same Peter–Weyl chain as Lemma 3.3, norm 1. Thus $V|p\rangle \supset -|\chi_1(p)\rangle$, the regeneration amplitude is $\langle p|V|\chi_1(p)\rangle = -1$, and the channel contributes $(-1)^2/(3-8) = -\frac{1}{5}$ on the diagonal.

5.3 Touching pairs, $p \neq q$ sharing one link

Two plaquettes sharing a link: inserting $\chi(U_r)$ with r supported on $p \cup q$ produces intermediate states on the 7 links of $p \cup q$ in which the shared link carries fused flux $s = 0$ or $s = 1$ ($V_{1/2} \otimes V_{1/2} = V_0 \oplus V_1$) and the other six links carry $j = \frac{1}{2}$. The two channels have energies $E_s = 6 \cdot \frac{3}{4} + s(s+1)$, i.e. $E_0 = \frac{9}{2}$, $E_1 = \frac{13}{2}$ (denominators $-\frac{3}{2}$, $-\frac{7}{2}$), with squared amplitudes $c_0^2 = \frac{1}{4}$, $c_1^2 = \frac{3}{4}$ (Appendix B, derived twice: $6j$ recoupling and Weingarten; $c_0^2 + c_1^2 = 1$ is the regeneration sum rule). With regeneration amplitudes $-c_s$, the touching-pair matrix element is

$$v \;=\; \frac{c_0^2}{3-E_0} + \frac{c_1^2}{3-E_1} \;=\; -\frac{2}{3}\cdot\frac{1}{4} - \frac{2}{7}\cdot\frac{3}{4} \;=\; -\frac{1}{6} - \frac{3}{14} \;=\; -\frac{8}{21}.$$

On the cubic lattice every bond lies in exactly four plaquettes (two planes through the bond, two sides of each), so each plaquette shares one link with $4 \times 3 = 12$ others and no two distinct plaquettes share two links; the touching adjacency A is 12-regular (machine census, Appendix C), and the touching channels contribute $v(A + 12I)$.

5.4 Disjoint pairs

If p, q are distinct and share no link, the intermediate state supported on $p \cup q$ (regeneration, $E_n = 6$, denominator -3) contributes $-\frac{1}{3}$ off-diagonal. The diagonal disjoint bookkeeping contributes $-\frac{|\mathcal{P}|-13}{3}$: the sum over q disjoint from p has $|\mathcal{P}| - 13$ terms (12 touching plus p itself excluded).

6 Effective Hamiltonian, vacuum subtraction, and the head of the band

Summing the channels, with A the touching adjacency and J all-ones,

$$M_2 \;=\; v(A + 12I) \;+\; \tfrac{1}{3}J \;-\; \tfrac{1}{5}I \;-\; \tfrac{1}{3}\big(J - I - A\big) \;-\; \tfrac{|\mathcal{P}|-13}{3}\,I \;=\; \Big(\tfrac{1}{3} + v\Big)A + c\,I, \qquad (2)$$

$c = \frac{1}{3} + 12v - \frac{1}{5} - \frac{|\mathcal{P}|-13}{3}$: the all-ones operator cancels *as an operator identity*. This identity is verified exactly at $L = 5$ ($|\mathcal{P}| = 375$) in [6], symbolically over the plaquette link structure.

6.1 Ground-state subtraction and the vacuum-mixing consistency

The ground energy is, by non-degenerate analytic perturbation theory with intermediate states $|p\rangle$ (amplitude -1, denominator $0-3$),

$$E_0(\mu) = \mu^2 \sum_{p\in\mathcal{P}} \frac{|\langle p|V|\Omega\rangle|^2}{0-3} + O(\mu^4) = -\frac{|\mathcal{P}|}{3}\,\mu^2 + O(\mu^4);$$

the third-order term vanishes identically: its Rayleigh–Schrödinger expression is a sum of $\langle\Omega|V|n\rangle\langle n|V|m\rangle\langle m|V|\Omega\rangle/((0-E_n)(0-E_m))$, and $\langle\Omega|V|a\rangle \neq 0$ forces a to be a single-plaquette state: in the Peter–Weyl flux basis, any flux link outside $\operatorname{supp} r$ leaves an unpaired $\bar{U}$-factor and any r-link without matching flux leaves an unpaired U-factor (Appendix A, rule (ii)), so $\operatorname{supp} a = \operatorname{supp} r$ with flux $\frac{1}{2}$ throughout and $a \propto |p\rangle$ by Lemma 3.2 (at $L=4$ a winding support never coincides with a plaquette support); the middle factor is then some $\langle p|V|q\rangle = 0$ (Lemma 4.1); hence the $O(\mu^4)$ bound. Equivalently, in the one-particle sector the uniform mode $|u\rangle = |\mathcal{P}|^{-1/2}\sum_p |p\rangle$ mixes with the vacuum through $\langle u|V|\Omega\rangle = -|\mathcal{P}|^{1/2}$; the 2×2 block $\begin{pmatrix} 0 & -\mu|\mathcal{P}|^{1/2} \\ -\mu|\mathcal{P}|^{1/2} & 3 \end{pmatrix}$ has upper eigenvalue $3 + \frac{|\mathcal{P}|}{3}\mu^2 + O(\mu^4)$ and lower eigenvalue $-\frac{|\mathcal{P}|}{3}\mu^2 + O(\mu^4)$. The gap is band-bottom minus $E_0(\mu)$; the $+\frac{|\mathcal{P}|}{3}$ entering the $k=0$ band edge through $\frac{1}{3}J$ in (2) is *exactly* the same vacuum-intermediate contribution, counted once in M_2 and cancelled once by the subtraction $-E_0'' = -\frac{|\mathcal{P}|}{3}$:

$$c_2 \;=\; \lambda_{\min}\big((\tfrac{1}{3}+v)A\big) + c + \frac{|\mathcal{P}|}{3} \;=\; 12\Big(\tfrac{1}{3}+v\Big) + \tfrac{1}{3} + 12v - \tfrac{1}{5} + \tfrac{13}{3} \;=\; 24v + \frac{127}{15}. \qquad (3)$$

Since $\frac{1}{3}+v = -\frac{1}{21} < 0$, the minimum of $(\frac{1}{3}+v)A$ over the band is $(\frac{1}{3}+v)\lambda_{\max}(A)$, attained at $k=0$.

Lemma 6.1 (band head)**.** *For every $L \geq 4$, $\lambda_{\max}(A(k)) \leq 12$ for all crystal momenta $k \in [-\pi,\pi]^3$, with equality iff $k=0$; $\lambda_{\max}(A(0)) = 12$ with the uniform orientation eigenvector.*

Proof. Fourier transform in the translations: the fiber $A(k)$ is a 3×3 matrix indexed by plaquette orientation (xy, xz, yz): $A_{xy,xy} = 2\cos k_x + 2\cos k_y$, etc., with off-diagonal $A_{xy,xz} = 4\cos(k_y/2)\cos(k_z/2)$ and cyclic. By Gershgorin, every eigenvalue is $\leq (2\cos k_x + 2\cos k_y) + |A_{xy,xz}| + |A_{xy,yz}| \leq 4+8 = 12$, and equality forces $\cos k_x = \cos k_y = 1$ and $|\cos(k_y/2)| = |\cos(k_z/2)| = 1$ etc., i.e. $k=0$ in the Brillouin zone. At $k=0$, $A(0)$ is the adjacency of the finite graph and the all-ones vector gives $\lambda = 12$ because A is 12-regular. □

The *uniqueness* statements — simplicity of the Perron root, rigidity of equality $\lambda_{\max}(A(k)) = 12 \Rightarrow k=0$, and the fact that the exact deformed band minimum sits at $k=0$ for all sufficiently small μ, with a strictly positive quadratic gap of leading coefficient $\frac{4}{63}$ to the next momentum — are proved in Appendix E (Perron–Frobenius plus exact curvature $-\frac{4}{3}|k|^2$; machine-checked in `code/verify_bandhead.py`).

Proof of Theorem 1.1. $L \geq 5$: Lemma 3.1 gives $\mathcal{M}$ = plaquette states, orthonormal (Lemma 3.3). First order vanishes (Lemma 4.1). Degenerate perturbation theory to second order is regular because the cluster $E=3$ is separated from the rest of the spectrum of H_E on the gauge-invariant space by $\frac{3}{2}$: the spectrum of H_E is $\sum_\ell j_\ell(j_\ell+1)$ over spin networks, and the vertex rules (Lemma F.3) plus bipartiteness of the cubic lattice force every closed network with

$E < \frac{9}{2}$ other than the vacuum to be a 4-cycle at spin $\frac{1}{2}$ (2-valent vertices force constant spin along loops, odd loops do not exist, and the smallest theta graph carries $E \geq 7$); the next level is the 6-loop $E = \frac{9}{2}$. Thus the cluster is isolated by a gap $\frac{3}{2}$ uniformly in L, and the band functions are $E_\alpha(\mu, k) = 3 + \mu^2 \lambda_\alpha(M_2(k)) + O(\mu^3)$ uniformly in k (analyticity, Section 2.3). By (2), (3) and Lemma 6.1, the bottom of the band minus $E_0(\mu)$ is $3 + \big(24v + \frac{127}{15}\big)\mu^2 + O(\mu^3)$ with $v = -\frac{8}{21}$: $24v + \frac{127}{15} = -\frac{192}{21} + \frac{127}{15} = -\frac{64}{7} + \frac{127}{15} = \frac{-960+889}{105} = -\frac{71}{105}$. The $|\mathcal{P}|$ dependence cancels identically in (3); hence the coefficient is L-independent for $L \geq 5$. □

7 Volume independence

The cancellation of $|\mathcal{P}|$ in (3) is exact, not asymptotic: the extensive vacuum-intermediate term $\frac{1}{3}|\mathcal{P}|$ on the uniform mode is cancelled by the extensive diagonal disjoint term $-\frac{|\mathcal{P}|-13}{3}$ plus the subtraction $+\frac{|\mathcal{P}|}{3}$. What remains, $24v + \frac{127}{15}$, involves only the local touching geometry, identical for all $L \geq 5$ (the $13 = 12 + 1$ touching census is L-independent; at $L = 4$ the winding sector changes the spectrum, Section 8). The machine verification at $L = 5$ builds the full 375×375 operator over $\mathbb{Q}$ and confirms (2) entry-by-entry [6].

8 The $L = 4$ winding sector

At $L = 4$ there are 192 plaquettes and 48 straight winding loops (3 directions $\times$ 4^2 transverse positions); the census is exact and machine-checked. Winding states $|w\rangle$ are orthonormal to plaquette states and to each other by Lemma 3.3. First order remains zero (center parity, Lemma 4.1).

Lemma 8.1 (winding decoupling)**.** $\langle p|M_2|w\rangle = 0$ *for every plaquette* p *and winding loop* w *at* $L = 4$*.*

Proof. First the geometric census: $|p \cap w| \leq 1$ for all pairs — the two parallel links of a plaquette share their longitudinal coordinate, the four links of a straight winding have distinct longitudinal coordinates (exact census over all 192×48 pairs in `verify_census.py`). Every intermediate contributing to $\langle p|M_2|w\rangle$ arises from $V|w\rangle$: the vacuum (killed by $\langle\Omega|V|w\rangle = 0$, support: no plaquette support equals a winding support), $|\chi_1(w)\rangle$ (energy 8), fused two-loop states on $w \cup r$ (energies $9/2, 13/2$), or products $\chi_r\chi_w$ (energy 6). Pairing with $\langle p|V$ inserts one further plaquette character χ_s, and covering all four winding links would require $|p \cap w| + |s \cap w| \geq 4$ while the census gives ≤ 2; some winding link is therefore left uncovered, carrying a single unpaired U-factor in the fused and product channels (rule A.1(ii)) or two U-factors in one symmetric spin-1 slot in the χ_1 channel (Lemma A.2). Every channel vanishes link by link. □

Thus the second-order block is diagonal in the plaquette/winding split. For winding states: 16 touching channels (each $= v$), no χ_1 self channel, no vacuum channel ($\langle w|V|\Omega\rangle$ contributes only through the same parity argument), disjoint diagonal $-\frac{|\mathcal{P}|-16}{3}$, giving

$$c_{2,w}^{\mathrm{diag}} = 16v + \frac{16}{3} = -\frac{16}{21} \approx -0.7619 < -\frac{71}{105} \approx -0.6762.$$

The winding chain block has zero diagonal, so the head of the winding band is $\leq -16/21$, strictly below $-71/105$: at $L = 4$ the gap is governed by the winding sector and Theorem 1.1 fails. The exact $c_2(L{=}4) = -\frac{16}{21} + \lambda_{\min}(B_w)$ requires the two-link chain recoupling of the block B_w, which we have not computed; we state the bound and name the open value rather than guess it.

9 Scope: fixed volume, uniformity, continuum

Theorem 1.1 is a fixed-L perturbative identity: μ_0 may depend on L, and the $O_L(\mu^3)$ remainder is not controlled uniformly in L. It does not imply $\Delta_{\text{unif}}(\mu) > 0$, nor anything about $a \to 0$. The distinction between (i) the finite-volume gap, (ii) the volume-uniform gap at fixed coupling, and (iii) the continuum gap is enforced throughout, following the campaign's model specification [**?**]. The relation of the coefficient to published strong-coupling series (Kogut–Sinclair–Susskind and successors) and the novelty classification are recorded in the literature ledger [7]; absence of an exact printed match is reported as "not located," not as proof of novelty. For honesty about the ingredients we note explicitly that the individual second-order channel values $-\frac{1}{3}$, $-\frac{1}{6}$, $-\frac{3}{14}$ appearing in Section 5 are published, in print, in the closest fully accessible relative of our calculation — the 2+1-dimensional strong-coupling static-potential analysis of Morningstar, whose staple classes carry exactly these numbers (the recoupling weights $\frac{1}{4}$, $\frac{3}{4}$ and denominators $-\frac{3}{2}, -\frac{7}{2}$ are dimension-independent); what is not located in the accessible literature is the assembled 3+1-dimensional one-glueball-band coefficient $-\frac{71}{105}$ with exact vacuum subtraction and the finite-volume band-head theorem. The novelty classification is NOVELTY UNRESOLVED: the single object-matched prior source, Kogut–Sinclair–Susskind, Nucl. Phys. **B114** (1976) 199–236, remains ACCESS-INCOMPLETE after eleven independent retrieval routes. The exact equivalence test is recorded for whoever obtains the text: convert their printed second-order gap, under $\mu = 2g^{-6}$ and energies in units $g^2/2$, to $\Delta_{\text{phys}} = (3/2)g^2 - (142/105)g^{-10}$; agreement yields KNOWN, a different converted value yields NOVELTY SUPPORTED. No originality is claimed here.

Acknowledgments

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[6] Verification code (exact-rational SymPy, no numerical fitting): `code/verify_anchors.py`, `verify_identity.py`, `verify_census.py`, `verify_bandhead.py`, accompanying this manuscript.

[7] Supplementary documents accompanying this manuscript (static copies, checksummed in `MANIFEST.sha256`): the primary-source access log for B114 (1976) with the recorded equivalence test, and the gap taxonomy (fixed-volume / volume-uniform / continuum) used in Section 9. The scope and novelty statements of Section 9 are reproduced in full in the body of this paper, so the manuscript is logically self-contained.

A Peter–Weyl index chains and vanishing lemmas

Write $\chi(g) = \operatorname{Tr} g$ (spin $\frac{1}{2}$ character). For a loop $C = (\ell_1, \dots, \ell_4)$, $\chi(U_C) = U_1{}^a{}_b\, U_2{}^b{}_c\, U_3{}^c{}_d\, U_4{}^d{}_a$ (summation over repeated indices).

Lemma A.1 (vanishing rules)**.** *Let F be a monomial in link matrix coefficients with per-link bidegree (p_ℓ, q_ℓ) (numbers of U-type and $\bar{U}$-type factors of link ℓ). Then $\int F\, dU = 0$ unless*

(i) $p_\ell - q_\ell$ is even *for every ℓ (single-link center parity: the flip $U_\ell \mapsto -U_\ell$ preserves Haar measure and multiplies F by $(-1)^{p_\ell - q_\ell}$); and*

(ii) the representation tensor product carries an invariant vector after all intertwiners are contracted; in particular a link of bidegree $(1,0)$ or $(0,1)$ forces vanishing, since $V_{1/2}$ has no invariant and $\int U^a{}_b\, dU = 0$.

These are the only rules used in this paper. (Neither is a complete criterion: unequal-degree integrals such as $\int UU$ can *be nonzero for* $\mathrm{SU}(2)$*, via ε-pairings — see Lemma A.2.)*

Lemma A.2 (symmetric-slot rule)**.** $\int U^a{}_b\, U^c{}_d\, dU = \frac{1}{2}\, \varepsilon^{ac}\varepsilon_{bd}$*; consequently, if a monomial has some link contributing two U-factors occupying a single symmetric (spin-1) matrix coefficient and no $\bar{U}$-factor on that link, the integral vanishes, because ε^{ac} annihilates the symmetric pair.*

Proof. The left side is an intertwiner in $\mathrm{Inv}(V \otimes V \otimes V \otimes V)$ unique up to scale, and $\frac{1}{2}\varepsilon^{ac}\varepsilon_{bd}$ is the invariant pairing; the annihilation statement is $\varepsilon^{ac} S_{ac} = 0$ for symmetric S. □

Lemma A.3 (diagonal Gram)**.** $\langle \chi(U_C), \chi(U_C) \rangle = 1$.

Proof. $\overline{\chi(U_C)} = \chi(U_C^\dagger) = \bar{U}_4{}^a{}_d \bar{U}_3{}^d{}_c \bar{U}_2{}^c{}_b \bar{U}_1{}^b{}_a$. Apply Peter–Weyl orthogonality link by link, $\int U^a{}_b\, \overline{U^c{}_d}\, dU = \frac{1}{2}\delta^a_d\, \delta^c_b$: each of the four links contributes $\frac{1}{2}$; each of the four successive index closures contributes a factor 2; the product is $2^{-4} \cdot 2^4 = 1$. □

Lemma A.4 (off-support vanishing)**.** *If $C \neq C'$ are closed loops of length 4 on $\mathbb{T}^3_L$, then* $\langle \chi(U_C), \chi(U_{C'}) \rangle = 0$.

Proof. Some $\ell \in C \setminus C'$; the integrand has bidegree $(1,0)$ on ℓ, so Lemma A.1(ii) applies. □

B Recoupling coefficients $c_0^2 = \frac{1}{4}$, $c_1^2 = \frac{3}{4}$

B.1 Method 1: $6j$/F-move calculus

Let $|s\rangle$, $s \in \{0, 1\}$, be the orthonormal s-channel basis of the intertwiner space $\mathrm{Inv}(V_{1/2}^{\otimes 4})$: pair links (ℓ_1, ℓ_2) to spin s and (ℓ_3, ℓ_4) to the same s. The regeneration state $\chi(U_p)\, \chi(U_q)$ for touching

p, q (shared link ℓ_1) is, in link language, the four-valent contraction with ℓ_1 appearing twice; changing from the plaquette pairing to the shared-link pairing is the F-move

$$F = \begin{pmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}, \qquad F_{ss'} = \sqrt{(2s+1)(2s'+1)} \begin{Bmatrix} \frac{1}{2} & \frac{1}{2} & s \\ \frac{1}{2} & \frac{1}{2} & s' \end{Bmatrix},$$

i.e. the $6j$ values $\{\frac{1}{2}\frac{1}{2}0; \frac{1}{2}\frac{1}{2}0\} = \frac{1}{2}$, $\{\frac{1}{2}\frac{1}{2}1; \frac{1}{2}\frac{1}{2}1\} = -\frac{1}{6}$, $\{\frac{1}{2}\frac{1}{2}0; \frac{1}{2}\frac{1}{2}1\} = \frac{1}{2}$ ($\pm$ signs convention-dependent; only squares enter). Hence the touching-regeneration state decomposes with amplitudes $c_0 = F_{00} = \frac{1}{2}$, $c_1 = F_{10} = \frac{\sqrt{3}}{2}$: $c_0^2 = \frac{1}{4}$, $c_1^2 = \frac{3}{4}$, $c_0^2 + c_1^2 = 1$ (unit regeneration norm, Lemma A.3). Energies: the s-channel intermediate has six links of flux $\frac{1}{2}$ and the shared link of flux s: $E_s = 6 \cdot \frac{3}{4} + s(s+1) \in \{\frac{9}{2}, \frac{13}{2}\}$, so $v = \frac{c_0^2}{3-9/2} + \frac{c_1^2}{3-13/2} = -\frac{8}{21}$.

B.2 Method 2: Weingarten integration

$\langle p|V\,\Pi_s\,V|q\rangle$ for touching p, q reduces to integrals of products of four matrix coefficients on the shared link, $\int U^a{}_b U^c{}_d \bar{U}^e{}_f \bar{U}^g{}_h \, dU$, evaluated by the SU(2) Weingarten rule $\int U \otimes \bar{U} \otimes U \otimes \bar{U} = \frac{1}{6}(\mathrm{id} - \frac{1}{2}\mathrm{flip}) + \frac{1}{3}(\mathrm{flip} - \frac{1}{2}\mathrm{id})$ (dimension formulas $\int P_{\mathrm{sym}} = \frac{3}{4}$, $\int P_{\mathrm{asym}} = \frac{1}{4}$). Both methods agree exactly; the machine check is `verify_anchors.py` and `verify_identity.py` [6].

B.3 χ_1 regeneration

Pointwise fusion $\chi_{1/2}(g)^2 = \chi_0(g) + \chi_1(g)$ gives $V|p\rangle \supset -\mathbf{1} - |\chi_1(p)\rangle$; $|\chi_1(p)\rangle$ carries pure flux $j = 1$ on the four links of p ($E_n = 8$) and has norm 1 by the index chain of Lemma A.3 with the spin-1 representation ($\int \chi_1 \bar{\chi}_1$-type contraction, same $2^{\pm 4}$ bookkeeping). Amplitude $\langle p|V|\chi_1(p)\rangle = -1$, channel $-1/5$.

C Adjacency census and the band-head lemma

Census. On $\mathbb{T}^3_L$ every bond lies in exactly four plaquettes (two planes through the bond direction, two transverse sides), hence each plaquette shares exactly one link with 4 (links) $\times$ 3 (partners) = 12 others, and distinct plaquettes never share two links. The touching adjacency A is 12-regular; both the degree and the $\{0, 1\}$ intersection property are verified exhaustively at $L = 5$ ($n = 375$) in `verify_identity.py`.

Fiber decomposition. Translations act freely on plaquettes of each of the three orientations (xy, xz, yz); Fourier transform along $\mathbb{Z}^3_L$ block-diagonalizes A into 3×3 fibers

$$A(k) = \begin{pmatrix} 2\cos k_x + 2\cos k_y & 4\cos\frac{k_y}{2}\cos\frac{k_z}{2} & 4\cos\frac{k_x}{2}\cos\frac{k_z}{2} \\ 4\cos\frac{k_y}{2}\cos\frac{k_z}{2} & 2\cos k_x + 2\cos k_z & 4\cos\frac{k_x}{2}\cos\frac{k_y}{2} \\ 4\cos\frac{k_x}{2}\cos\frac{k_z}{2} & 4\cos\frac{k_x}{2}\cos\frac{k_y}{2} & 2\cos k_y + 2\cos k_z \end{pmatrix},$$

(derivation: the two bonds through which an xy-plaquette at the origin touches xz-plaquettes are displaced by $\pm e_y$ and $\pm e_z$ relative to the shared-bond identification, producing the product-of-cosines entries; each entry is the sum of the four phase weights of the corresponding touching family).

Gershgorin bound. Row xy: $\lambda \le 2\cos k_x + 2\cos k_y + 4|\cos\frac{k_y}{2}\cos\frac{k_z}{2}| + 4|\cos\frac{k_x}{2}\cos\frac{k_z}{2}| \le 4 + 8 = 12$; equality forces $\cos k_x = \cos k_y = 1$ and $|\cos\frac{k_{y,z}}{2}| = 1$, i.e. $k = 0$ in $[-\pi, \pi]^3$, where $A(0)\mathbf{1} = 12\,\mathbf{1}$ by regularity. This is Lemma 6.1 records the *upper bound*; uniqueness of the maximizer and of the exact band head are proved in Appendix E, without numerics.

D Normalization dictionary and reproducibility

Dictionary. The theorems are statements about $H(\mu) = \sum_\ell C_2(\ell) + \mu \sum_p (2 - \Re \operatorname{Tr} U_p)$. For $H_{\mathrm{KS}} = \frac{g^2}{2} \sum E^2 - \frac{1}{2g^4} \sum_p (\operatorname{Tr} U_p + \operatorname{Tr} U_p^\dagger)$ at $a = 1$: $\mu = 2g^{-6}$, physical energy $= \frac{g^2}{2} \times$(dimensionless). If an alternative convention rescales the expansion parameter by a constant, $\mu' = c\mu$, the theorem reads $\Delta = 3 - \frac{71}{105\, c^2} \mu'^2 + O_L(\mu'^3)$. Conventions quoting $\mu' = g^{-4}$ arise from applying the same dimensionless form to $\widetilde{H} = \frac{2}{g^2} H_{\mathrm{KS}}$ with g reparametrized so that the plaquette coefficient reads $-\mu' \sum_p \Re \operatorname{Tr} U_p$ with $\mu' = g^{-4}$ (a different overall normalization of H_{KS}); the dictionary must then be applied to the full dimensionless Hamiltonian, not to the symbol μ alone — convert the Hamiltonian, not just the coupling.

Reproducibility. `code/verify_anchors.py` proves $\|\chi(U_p)\|^2 = 1$, $\langle p|V|\Omega\rangle = -1$, $\langle p|V|q\rangle = 0$, and the c_s^2 by two independent exact methods (Weyl integration and Weingarten); `code/verify_identity.py` builds the 375×375 operator at $L = 5$ over $\mathbb{Q}$, verifies $M_2 = (\frac{1}{3} + v)A + cI$ entry-by-entry, the 2×2 mixing consistency $\langle u|\frac{1}{3}J|u\rangle = |\mathcal{P}|/3$, and $c_2 = -71/105$; `code/verify_census.py` runs the $L = 4$ census $192 + 48 = 240$ and the winding touching count. All computations are exact rational arithmetic (SymPy); no numerical fitting, no floating-point assumptions.

E Band-head uniqueness: Perron–Frobenius and curvature

Lemma E.1 (maximal eigenvalue, strict form)**.** *For every $L \geq 4$ and every $k \in [-\pi, \pi)^3$, $\lambda_{\max}(A(k)) \leq 12$, with equality* if and only if *$k = 0$; moreover $\lambda_{\max}(A(0)) = 12$ is* simple *and its eigenline is spanned by the uniform orientation vector.*

Proof. (1) Comparison. For a Hermitian matrix B, writing $|B|$ for the entrywise absolute value, $\lambda_{\max}(B) = \max_{\|x\|=1} \langle x, Bx\rangle \leq \max_{\|x\|=1} |\langle x, Bx\rangle| \leq \max_{\|y\|=1,\, y \geq 0} \langle y, |B|y\rangle = \rho(|B|)$, the last step being the Collatz–Wielandt characterization of the Perron root of a nonnegative matrix. Applied to $B = A(k)$: every entry of $|A(k)|$ is bounded by the corresponding entry of $A(0)$ (diagonals: $|2\cos k_x + 2\cos k_y| \leq 4$; off-diagonals: products of cosines bounded by 4), and the spectral radius is monotone on nonnegative matrices, so $\lambda_{\max}(A(k)) \leq \rho(|A(k)|) \leq \rho(A(0))$.

(2) Irreducibility. The touching graph of plaquettes on $\mathbb{T}^3_L$ is connected: in one plane the plaquettes form the $L \times L$ grid under link-sharing, and the three plane-families meet at every site. Hence $A(0)$ is irreducible; 12-regularity gives $A(0)\mathbf{1} = 12\mathbf{1}$, so $\rho(A(0)) = 12$, and Perron–Frobenius makes 12 a simple eigenvalue with strictly positive eigenvector; regularity then forces that eigenvector to be $\mathbf{1}$.

(3) Rigidity of equality. Suppose $\lambda_{\max}(A(k)) = 12$. Then equality holds throughout (1), in particular $\rho(|A(k)|) = \rho(A(0))$. If some entry of $|A(k)|$ were strictly smaller than that of $A(0)$, then, $A(0)$ being irreducible, the strict monotonicity of the Perron root (add a nonnegative perturbation supported on that entry and use the Perron eigenvector of $|A(k)|$, which is positive on the irreducible support) gives $\rho(|A(k)|) < \rho(A(0))$, contradiction. Hence $|A(k)| = A(0)$ entrywise. The off-diagonal equalities give $|\cos \frac{k_i}{2}| = 1$ for each i; for $k_i \in [-\pi, \pi)$ this forces $k_i = 0$ (the diagonal equalities then hold automatically). □

Proposition E.2 (exact band head for small μ)**.** *Let $L \geq 5$. There are $\mu_1(L) > 0$ and $\kappa(L) > 0$ such that for $0 < \mu < \mu_1(L)$ the minimum of the deformed $E = 3$ band over the discrete momentum lattice $k \in \frac{2\pi}{L}\mathbb{Z}^3 \cap [-\pi, \pi)^3$ is attained uniquely at $k = 0$, in the uniform orientation mode, and the gap to the next band minimum is at least $\kappa(L)\mu^2$.*

Proof. By Lemma E.1(2) the eigenvalue 12 of $A(0)$ is simple, so $\lambda_{\max}(A(k))$ is real-analytic near $k = 0$ (analytic perturbation of a simple eigenvalue of the 3×3 fiber). Differentiating the characteristic polynomial $p(k, \lambda) = \det(\lambda - A(k))$ at $(0, 12)$: with $A(k) = A(0) + A_2(k) + O(|k|^4)$, A_2 collecting the quadratic terms (diag entries $-k_x^2 - k_y^2$ etc., off-diagonal entries $-\frac{1}{2}(k_y^2 + k_z^2)$ etc.), first-order perturbation theory on the simple eigenpair $(12, \mathbf{1}/\sqrt{3})$ gives $\lambda_{\max}(A(k)) = 12 + \langle \mathbf{1}, A_2(k)\mathbf{1} \rangle / 3 + O(|k|^4) = 12 - \frac{4}{3}|k|^2 + O(|k|^4)$: the diagonal part contributes $-\frac{1}{3} \cdot 2|k|^2$ and the off-diagonal part $-\frac{1}{3} \cdot 2|k|^2$, summing to the isotropic $-\frac{4}{3}|k|^2$ (exact rational; the cubic term vanishes by the inversion symmetry $k \mapsto -k$ of the fiber; verified in `code/verify_bandhead.py`). Since $\frac{1}{3} + v = -\frac{1}{21} < 0$, the lowest band function obeys, uniformly in k (Analyticity subsection of Sec. 2), $E_{\min}(\mu, k) = 3 + \mu^2[(\frac{1}{3} + v)\lambda_{\max}(A(k)) + c] + O(\mu^3)$ with the $O(\mu^3)$ constant depending only on L (finite lattice, bounded perturbation). Hence for $k \neq 0$, $E_{\min}(\mu, k) - E_{\min}(\mu, 0) = \mu^2 \cdot \frac{1}{21} \cdot \frac{4}{3}|k|^2 + O(\mu^2|k|^4) + O(\mu^3) = \frac{4}{63}\mu^2|k|^2 + O(\mu^2|k|^4) + O(\mu^3)$. On the discrete lattice the smallest nonzero $|k|$ is $2\pi/L$, so choosing $\mu_1(L)$ with $C_{\mathrm{rem}}\, \mu_1 < \frac{4}{63}(2\pi/L)^2$ makes the right side strictly positive for every $k \neq 0$, with margin $\kappa(L)\mu^2$. Uniqueness of the maximizer at $k = 0$ is Lemma E.1(2). □

F Exhaustion of second-order intermediate states

The second-order effective operator on the $E = 3$ eigenspace is $M_2 = -P\, H_M\, R_0\, H_M\, P$ restricted to the cluster with the vacuum channel separated, R_0 the reduced resolvent of H_E at 3 on $\mathcal{M}\ominus$ vacuum directions; every claim about M_2 below is about matrix elements of this reduced operator between plaquette states. The intermediate-state enumeration in Proposition F.4 is stated for the *unreduced* second-order expression $P\, H_M\, (H_E - 3)^{-1} H_M\, P$ before vacuum subtraction; channel (i) there is precisely the term removed by the subtraction and reappears as the normalization correction of Section 5.1.

Lemma F.1 (intermediate selection)**.** *If $\langle n|H_M|p\rangle \neq 0$ for a plaquette state $|p\rangle$ and a gauge-invariant eigenstate $|n\rangle$ of H_E, then for some plaquette q, $|n\rangle$ has an index chain supported in $\mathrm{supp}(p) \cup \mathrm{supp}(q)$ in which every link of $\mathrm{supp}(q) \setminus \mathrm{supp}(p)$ carries spin $\frac{1}{2}$, every link of $\mathrm{supp}(p) \cap \mathrm{supp}(q)$ carries spin 0 or 1, and every link of $\mathrm{supp}(p) \setminus \mathrm{supp}(q)$ carries spin $\frac{1}{2}$. In particular $E(n) = \sum j(j+1)$ over that support.*

Proof. $\langle n|H_M|p\rangle = \sum_q \int \overline{\chi_1(n)}\, \chi_{1/2}(U_q)\, \chi_1(p)\, dU$. For each link ℓ, the two plaquette characters $\chi_1(p)\, \chi_{1/2}(U_q)$ carry, in their tensor-network expansion, two fundamental slots around the closed spin-1 chain of p and one fundamental slot around the chain of q. A link in $\mathrm{supp}(q) \setminus \mathrm{supp}(p)$ carries exactly one fundamental pair; degree matching of the U_ℓ integral against the conjugate spin-$j_\ell(n)$ chain of $\overline{\chi_1(n)}$ then forces $j_\ell(n) = \frac{1}{2}$. A link in $\mathrm{supp}(p) \cap \mathrm{supp}(q)$ carries three slots (odd); the admissible intermediate spins there are the recoupling channels of the two fundamental legs of q against the spin-1 chain of p computed in Appendices A–B: the singlet channel ($s = 0$, weight $c_0^2 = \frac{1}{4}$) and the triplet channel ($s = 1$, weight $c_1^2 = \frac{3}{4}$), so $j_\ell(n) \in \{0, 1\}$ on the overlap. **Open (revision item R3'):** the exhaustive form of this selection rule — that the recoupling channels of Appendices A–B are the *only* nonvanishing overlap contributions, with no higher-spin chain terms — is currently established by the exact full-space operator identity of Appendix D and the complete vertex-rule enumeration of Proposition F.4, not by a standalone link-level parity argument; a self-contained derivation is required before external review of the exhaustion proof. Rule (ii) then requires an intertwiner at each site of the union support; sites outside

$\mathrm{supp}(p) \cup \mathrm{supp}(q)$ carry unmatched single $\bar{U}$ factors from $\overline{\chi_1(n)}$ unless $j_\ell(n) = 0$ there, giving the support claim. □

Lemma F.2 (geometric trichotomy)**.** *For plaquettes $p \neq q$ on $\mathbb{T}^3_L$, $|\mathrm{supp}(p) \cap \mathrm{supp}(q)| \in \{0, 1\}$: distinct plaquettes share no link or exactly one link, never two. Hence the pairs (p, q) contributing to $\langle p|M_2|p'\rangle$ fall into exactly three geometric classes: $q = p$; q touching p (one shared link); q disjoint from p.*

Proof. Two distinct unit squares of the cubic lattice sharing two edges would share two adjacent edges, forcing a common vertex and two common edge directions, hence a common plane and a common third and fourth vertex — i.e. the same square; or share two opposite edges, forcing the same four vertices and the same plane. Both give $p = q$. On $\mathbb{T}^3_L$ with $L \geq 4$ the projection is injective on unit squares except for the winding loops of Lemma (winding census), which are excluded from the plaquette set by definition ($4 < L$ edges). Exhaustive verification at $L = 5$ (375 plaquettes, all pairs) is in `verify_identity.py`. □

Lemma F.3 (vertex singlet rules)**.** *For $j_1, j_2 \in \frac{1}{2}\mathbb{N}$: $\dim \mathrm{Inv}(j_1 \otimes j_2) = 1$ if $j_1 = j_2$ and 0 otherwise. For three spins, $\dim \mathrm{Inv}(j_1 \otimes j_2 \otimes j_3) = 1$ if $|j_1 - j_2| \leq j_3 \leq j_1 + j_2$ and $j_1 + j_2 + j_3 \in \mathbb{Z}$, and 0 otherwise.*

Proof. Clebsch–Gordan: $j_1 \otimes j_2 \simeq \bigoplus_{j=|j_1-j_2|}^{j_1+j_2} j$ (multiplicity-free); Inv picks the spin-0 summand. Two spins: 0 appears iff $j_1 = j_2$. Three: fuse $j_1 \otimes j_2$ and pick j_3 in the range; the parity condition is $j_1 + j_2 + j_3 \in \mathbb{Z}$. □

Proposition F.4 (channel exhaustion)**.** *The intermediate states with nonzero contribution to the unreduced second-order sum before vacuum subtraction are exhausted by exactly the four channels computed in Sections 5.1–5.4. Channel (i) is carried entirely by the normalization (vacuum-subtraction) term of Section 5.1 and does not enter M_2; channels (ii)–(iv) exhaust the intermediates of the reduced operator $\langle p|M_2|p'\rangle$:* (i) *the vacuum intermediate, value $-\frac{1}{3}$ per p (normalization only);* (ii) *the same-plaquette spin-1 regeneration, $E(n) = 8$, value $-\frac{1}{5}$;* (iii) *touching pairs, two intermediate families with shared-link spin 0 ($E = \frac{9}{2}$) and spin 1 ($E = \frac{13}{2}$), each with a unique intertwiner, combining through the Appendix-B weights $c_0^2 = \frac{1}{4}$, $c_1^2 = \frac{3}{4}$ into $v = -\frac{8}{21}$;* (iv) *disjoint pairs, unique intermediate with all eight links at spin $\frac{1}{2}$, $E = 6$, value $-\frac{1}{3}$. Every admissible intermediate below has intertwiner multiplicity exactly 1, so no multiplicity factors are hidden.*

Proof. By Lemma F.1 any intermediate chain is supported on $\mathrm{supp}(p) \cup \mathrm{supp}(q)$ with spins in $\{0, 1\}$ on $\mathrm{supp}(p) \cap \mathrm{supp}(q)$ and $\frac{1}{2}$ elsewhere; by Lemma F.2 the pair geometry is one of three cases, and the admissible assignments are exactly those whose every vertex passes Lemma F.3. This is a finite check and it is executed exactly (integer arithmetic) in `code/verify_intertwiners.py`: *self* ($q = p$): 16 assignments on the 4-cycle; the four vertices are 2-valent in the support, so Lemma F.3 forces adjacent spins equal, and going around the cycle leaves exactly two admissible assignments — all-0 ($E = 0$, channel (i)) and all-1 ($E = 8$, channel (ii)); every mixed 0/1 assignment fails at a vertex of unequal adjacent spins. *Touching*: the shared link carries $s \in \{0, 1\}$; the six singly covered links carry $\frac{1}{2}$; the four non-shared vertices are 2-valent with equal spins (pass), and the two endpoints of the shared link are 3-valent with $(s, \frac{1}{2}, \frac{1}{2})$, admissible for both $s = 0$ and $s = 1$ with multiplicity one, giving $E = \frac{9}{2}$ and $E = \frac{13}{2}$ with denominators $-\frac{3}{2}$ and $-\frac{7}{2}$. The shared link enters through the two singlet channels of the two fundamental legs at each endpoint with the squared recoupling weights $c_0^2 = \frac{1}{4}$, $c_1^2 = \frac{3}{4}$ of Appendix B, so the

summed touching value is $v = \frac{1/4}{-3/2} + \frac{3/4}{-7/2} = -\frac{1}{6} - \frac{3}{14} = -\frac{8}{21}$, matching Section 5.3 exactly (machine-checked). *Disjoint*: both squares regenerate the all-$\frac{1}{2}$ network (unique invariant per square by the 2-valent rule), $E = 6$, denominator -3, value $-\frac{1}{3}$, matching Section 5.4 (machine-checked). Multiplicities: 1 (self), 12 (touching, 12-regularity), $|\mathcal{P}| - 13$ (disjoint), matching (2). No other pair geometry exists (Lemma F.2) and no other spin assignment on these supports passes Lemma F.3, so the list is exhaustive. □